YEAR 8 MATHEMATICS · GETTING READY FOR THE YEARLY EXAM

Catch-up revision booklet

Pythagoras' theorem · Fractions, decimals and percentages · Algebraic techniques

Name: _____

Date started: _____

1 LEARN IT read it · **2 WORKED EXAMPLE** follow each line · **3 YOUR TURN** fill the blanks · **4 CHECK** with the answer booklet

1 Pythagoras' theorem

NSW syllabus: **MA4-PYT-C-01** applies Pythagoras' theorem to solve problems in various contexts · **MA4-IND-C-01** squares and square roots · **MAO-WM-01** working mathematically

Yearly exam
Section C · Q15, Q17, Q18

1A Squares, powers and square roots

EASY · support & core · EXAM Q17

LEARN IT

- **Squaring** multiplies a number by itself: $7^2 = 7 \times 7 = 49$ (not 7×2).
- A **power** says how many times to multiply: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.
- The **square root** undoes squaring: $\sqrt{49} = 7$, because $7^2 = 49$.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

← perfect squares: learn these by heart

YOUR TURN 1

Evaluate.

a $6^2 =$ _____ **b** $11^2 =$ _____ **c** $0.3^2 =$ _____ **d** $3^4 =$ _____ **e** $10^3 =$ _____

f $\sqrt{81} =$ _____ **g** $\sqrt{196} =$ _____ **h** $5^2 + 12^2 =$ _____ **i** $\sqrt{5^2 + 12^2} =$ _____

1B The hypotenuse and Pythagoras' theorem

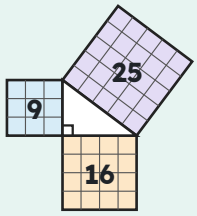
EASY · core

LEARN IT

- A **right-angled triangle** has one 90° angle, marked with a small square.
- The **hypotenuse** is the **longest side**. It is **opposite** (across from) the right angle.
- **Pythagoras' theorem:** in a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the two shorter sides.

$c^2 = a^2 + b^2$ where c is the hypotenuse and a and b are the two shorter sides.

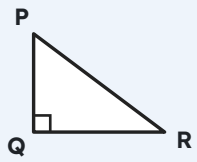
See it: in a 3, 4, 5 triangle, the squares on the shorter sides (9 + 16) match the square on the hypotenuse (25).



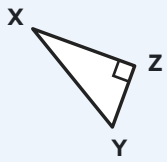
YOUR TURN 2

Name the hypotenuse in a, b and c. In d, write Pythagoras' theorem for the triangle.

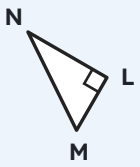
a Hypotenuse: _____



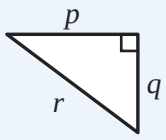
b Hypotenuse: _____



c Hypotenuse: _____



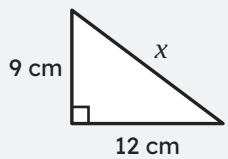
d Rule: _____



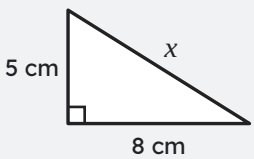
1C Finding the hypotenuse

MEDIUM · core · EXAM Q15a

WORKED EXAMPLE



$c^2 = a^2 + b^2$ ← write the rule
$x^2 = 9^2 + 12^2$ ← substitute
$x^2 = 81 + 144 = 225$
$x = \sqrt{225} = \mathbf{15\ cm}$ ← square root

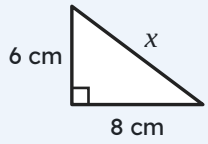


$x^2 = 5^2 + 8^2 = 25 + 64 = 89$
$x = \sqrt{89}$ ← exact answer (a surd)
$x \approx 9.433\,98\ldots \approx \mathbf{9.4\ cm}$ (1 d.p.)
$\approx$ means "is approximately equal to".

YOUR TURN 3

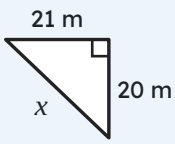
Find the length of the hypotenuse, x. Grey steps are done for you.

a Fill in the blanks.



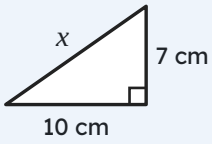
$x^2 = 6^2 + 8^2$
$x^2 =$ _____ + _____
$x^2 =$ _____
$x = \sqrt{\text{_____}} =$ _____ cm

b Fill in the blanks.



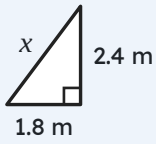
$x^2 =$ _____^{2} + _____^{2}
$x^2 =$ _____ + _____ = _____
$x =$ _____ m

c Round to 1 decimal place.



$x \approx$ _____ cm

d Show your working.



$x =$ _____ m

1D Finding a shorter side

MEDIUM · core

LEARN IT

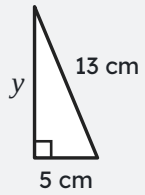
Which side is missing?
Find the right angle. The side opposite it is the hypotenuse.

Missing the hypotenuse → ADD
$c^2 = a^2 + b^2$

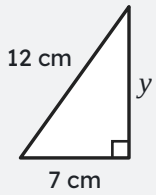
Missing a shorter side → SUBTRACT
$a^2 = c^2 - b^2$ (hypotenuse² − the other side²)

Check: a shorter side must come out **smaller** than the hypotenuse.

WORKED EXAMPLE



$y^2 = 13^2 - 5^2$ ← subtract
$y^2 = 169 - 25 = 144$
$y = \sqrt{144}$ = **12 cm**
Check: 12 is less than 13 ✓

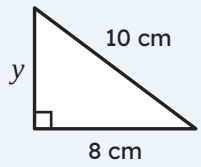


$y^2 = 12^2 - 7^2$
$y^2 = 144 - 49 = 95$
$y = \sqrt{95} \approx 9.746\ldots \approx$ **9.7 cm**
Check: 9.7 is less than 12 ✓

YOUR TURN 4

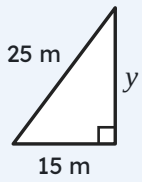
Find the length of the unknown side. Round to 1 decimal place where needed.

a Fill in the blanks.



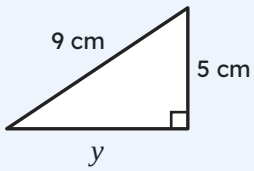
$y^2 = 10^2 - 8^2$
$y^2 =$ ____ − ____
$y^2 =$ ____
$y = \sqrt{\text{____}} =$ ____ cm

b Fill in the blanks.



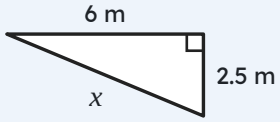
$y^2 =$ ____² − ____²
$y^2 =$ ____ − ____ = ____
$y =$ ____ m

c Round to 1 decimal place.



$y \approx$ ____ cm

d **Add or subtract?** Find x.



$x =$ ____ m

1E Surds: exact answers

MEDIUM · core EXAM Q18a

LEARN IT

- $\sqrt{81} = 9$ is a whole number, so $\sqrt{81}$ is **not** a surd.
- $\sqrt{89} = 9.433\ 981\ 1\ldots$ goes on forever and never repeats, so $\sqrt{89}$ **is** a surd.
- The square root of a whole number is a **surd** when the number is **not** a perfect square.
- **Exact** answer: $\sqrt{89}$. **Rounded** answer: ≈ 9.4

YOUR TURN 5

a Circle the surds: $\sqrt{1}$ $\sqrt{8}$ $\sqrt{30}$ $\sqrt{49}$ $\sqrt{45}$ $\sqrt{100}$ $\sqrt{2}$ $\sqrt{144}$

b $\sqrt{30}$ lies between which two whole numbers? Explain.

_____

c A right-angled triangle has shorter sides 3 cm and 5 cm. Find the hypotenuse.

Exact: $x =$ ____ cm Rounded: $x \approx$ ____ cm

1F Pythagoras in real life

CHALLENGING · core, exam style

LEARN IT

Four steps for worded problems

1. Draw a diagram. Mark the right angle.
2. Label the sides you know; use a pronumeral for the unknown.
3. Unknown hypotenuse → add. Shorter side → subtract.
4. Answer in a sentence, with units.

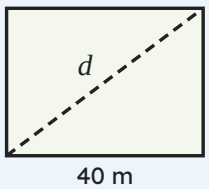
WORKED EXAMPLE



A 4.1 m ladder leans against a wall. Its foot is 0.9 m from the wall. How high up the wall does it reach?
The ladder is the hypotenuse, so subtract:
$h^2 = 4.1^2 - 0.9^2 = 16.81 - 0.81 = 16$
$h = \sqrt{16} = 4$. **The ladder reaches 4 m up the wall.**

YOUR TURN 6

a A rectangular paddock is 40 m long and 30 m wide. How long is the diagonal path across it?



The path is ____ m long.

b A phone screen is 7 cm wide and 15 cm tall. Find the length of its diagonal, to 1 decimal place. Draw a diagram first.

c Ava flies a kite on a 30 m string pulled tight. The kite is directly above a point 18 m from Ava's hand. How high is the kite above her hand?

1G Is it a right-angled triangle? (the converse of Pythagoras' theorem)

MEDIUM · core · EXAM Q15b

LEARN IT

1. Square the **longest** side.
2. Square the two shorter sides and **add**.
3. If the answers are **equal**, the triangle **is** right-angled. The right angle is opposite the longest side. If they are not equal, it is **not** right-angled.

Always finish with a sentence that answers the question.

WORKED EXAMPLE

(a) 12 cm, 16 cm, 20 cm: $20^2 = 400$ and $12^2 + 16^2 = 144 + 256 = 400$

$400 = 400$, so **the triangle is right-angled.**

(b) 5 cm, 7 cm, 9 cm: $9^2 = 81$ and $5^2 + 7^2 = 25 + 49 = 74$

$81 \neq 74$, so **the triangle is not right-angled.**

YOUR TURN 7

Is each triangle right-angled? The sides are in centimetres.

Sides	Longest side²	Sum of the squares of the other two sides	Equal?	Right-angled?
a 6, 8, 10	$10^2 =$ ____	$6^2 + 8^2 =$ ____ + ____ = ____	Yes / No	Yes / No
b 4, 5, 6				
c 9, 40, 41				
d 10, 11, 15				
e 1.5, 2, 2.5				
f 11, 60, 61				

1H Pythagorean triads

MEDIUM · core

LEARN IT

A **Pythagorean triad** is three whole numbers that fit $a^2 + b^2 = c^2$, such as 3, 4, 5 $(9 + 16 = 25)$.

Multiply a triad by any whole number to get another: 3, 4, 5 → 6, 8, 10 → 9, 12, 15.

YOUR TURN 8

a Show that 5, 12, 13 is a Pythagorean triad.

b Multiply 3, 4, 5 by 5. Check that the new numbers make a triad.

c Complete each triad.

9, 12, ____

10, ____, 26

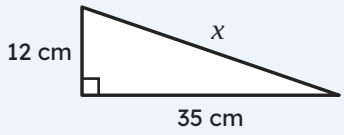
1I Exam-style practice

CHALLENGING · core, exam style · EXAM Q15

YOUR TURN 9

Show all working.

1 Find the length of the hypotenuse, x. 2 marks

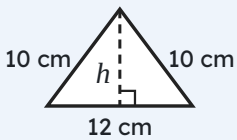


$x =$ ____ cm

2 A triangle has sides 8 cm, 10 cm and 13 cm. Is it a right-angled triangle? Show working. 2 marks

3 A builder checks that a corner is square. She marks 60 cm along one wall and 80 cm along the other. The distance between the marks is 1 m. Is the corner a right angle? 2 marks

4 An isosceles triangle has two sides of 10 cm and a base of 12 cm. Find its perpendicular height, h. 2 marks



$h =$ ____ cm

EXTENSION · STAGE 5 PATH CONTENT · YOUR EXAM ASKS THIS IN Q18b

Simplifying a surd: find the **largest square factor** (4, 9, 16, 25, 36, …) and take its square root out.

$\sqrt{18} = \sqrt{9} \times 2 = \sqrt{9} \times \sqrt{2} = \mathbf{3\sqrt{2}}$ $\sqrt{48} = \sqrt{16} \times 3 = \mathbf{4\sqrt{3}}$ (not $\sqrt{4} \times 12$: 4 is not the largest square factor)

Simplify.

a $\sqrt{8}$ = ____ **b** $\sqrt{20}$ = ____ **c** $\sqrt{27}$ = ____ **d** $\sqrt{75}$ = ____

I can do this on my own: ☐ hypotenuse ☐ shorter side ☐ right angle test ☐ surds

Check your answers in the answer booklet. Diagrams are drawn to scale unless marked "not to scale".

2 Fractions, decimals and percentages

NSW syllabus: **MA4-FRC-C-01** represents and operates with fractions, decimals and percentages to solve problems · **MAO-WM-01** working mathematically

Yearly exam
Section C · Q16 · Section B · Q9

2A Equivalent fractions, simplifying and mixed numerals

EASY · support & core

LEARN IT

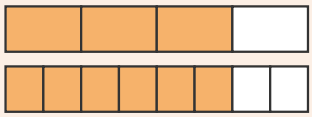
Equivalent fractions — Multiply or divide the **numerator** (top) and **denominator** (bottom) by the same number.

Simplify — Divide the top and bottom by their **highest common factor** (HCF): $\frac{18}{30} = \frac{18 \div 6}{30 \div 6} = \frac{3}{5}$

Mixed → improper — $2\frac{3}{5} = \frac{2 \times 5 + 3}{5} = \frac{13}{5}$

Improper → mixed — $\frac{17}{4}$: $17 \div 4 = 4$ remainder 1, so $\frac{17}{4} = 4\frac{1}{4}$

SEE IT



$\frac{3}{4}$

$\frac{6}{8}$

Same amount shaded, so $\frac{3}{4} = \frac{6}{8}$. Cutting each quarter in half doubles the top **and** the bottom.

YOUR TURN 1

a Complete: $\frac{2}{3} = \frac{\square}{12}$

b Simplify $\frac{15}{25} =$ _____

c Simplify $\frac{24}{32} =$ _____

d Improper fraction: $3\frac{2}{7} =$ _____

f Order from smallest to largest: $\frac{2}{3}$ $\frac{3}{4}$ $\frac{5}{8}$

_____

e Mixed numeral: $\frac{29}{6} =$ _____

2B Adding and subtracting fractions

MEDIUM · core

LEARN IT

Fractions need the **same denominator** before you add or subtract.

1. Find the **lowest common denominator** (LCD) and rewrite each fraction.
2. Add or subtract the numerators. Keep the denominator.
3. Mixed numerals: change to improper fractions first.

WORKED EXAMPLE

$\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \mathbf{\frac{11}{12}}$ (LCD of 3 and 4 is 12)

$3\frac{1}{2} - 1\frac{2}{3} = \frac{7}{2} - \frac{5}{3} = \frac{21}{6} - \frac{10}{6} = \frac{11}{6} = \mathbf{1\frac{5}{6}}$

YOUR TURN 2

Evaluate. Write answers in simplest form.

a $\frac{1}{2} + \frac{1}{3}$

$= \frac{\square}{6} + \frac{\square}{6} = \frac{\square}{6}$

b $\frac{4}{5} - \frac{1}{2}$

$= \frac{\square}{10} - \frac{\square}{10} = \frac{\square}{10}$

c $\frac{3}{4} + \frac{5}{6}$

$= \frac{\square}{12} + \frac{\square}{12} =$ _____

d $2\frac{1}{4} + 1\frac{2}{3}$

_____

$=$ _____

e $5 - 2\frac{3}{8}$

_____

$=$ _____

2C Multiplying and dividing fractions

MEDIUM · core

LEARN IT

× Multiply the tops, multiply the bottoms. Cancel first to keep numbers small.

÷ **Keep** the first fraction, **change** ÷ to ×, **flip** the second.

Mixed numerals → improper fractions first. "**of**" means ×: $\frac{2}{3}$ of 45 = 45 ÷ 3 × 2 = 30.

WORKED EXAMPLE

$\frac{4}{9} \times \frac{3}{8} = \frac{\cancel{4}^{1}}{\cancel{9}_{3}} \times \frac{\cancel{3}^{1}}{\cancel{8}_{2}} = \mathbf{\frac{1}{6}}$ (4 and 8 share 4; 3 and 9 share 3)

$2\frac{1}{2} \div \frac{3}{4} = \frac{5}{2} \times \frac{4}{3} = \frac{20}{6} = \frac{10}{3} = \mathbf{3\frac{1}{3}}$

YOUR TURN 3

Evaluate. Write answers in simplest form.

a $\frac{2}{5} \times \frac{3}{7}$

$=$ _____

b $\frac{5}{6} \times \frac{3}{10}$

$=$ _____

c $1\frac{1}{3} \times \frac{3}{8}$

_____

$=$ _____

d $\frac{3}{4} \div \frac{1}{2}$

$= \frac{3}{4} \times \frac{\square}{\square} =$ _____

e $\frac{2}{3} \div 4$

_____

$=$ _____

f $3\frac{1}{2} \div 1\frac{3}{4}$

_____

$=$ _____

g A farmer has 36 sheep. One-third are black, one-quarter are brown and the rest are white. How many are white?

_____

h A cake needs $2\frac{1}{4}$ cups of flour. How much flour is needed for 3 cakes?

_____

2D Fractions ↔ decimals, and rounding

EASY · support & core

LEARN IT

- **Decimal → fraction:** use place value, then simplify: $0.35 = \frac{35}{100} = \frac{7}{20}$
- **Fraction → decimal:** divide the top by the bottom: $\frac{3}{8}$ = $3 \div 8 = 0.375$ (it stops: **terminating**)
- Some decimals repeat forever (**recurring**). Dots go over the first and last digits that repeat: $\frac{2}{3} = 0.666\ldots = 0.\dot{6}$ $\frac{5}{11} = 0.4545\ldots = 0.\dot{4}\dot{5}$

LEARN IT

Rounding to 2 decimal places: look at the **3rd** decimal place.
5 or more → round up. 4 or less → keep the digit.
2.7368 → **2.74** 2.7328 → **2.73**
"Nearest tenth" means 1 decimal place.

YOUR TURN 4

a 0.8 as a fraction
= _____

b 0.45 as a fraction
= _____

c 0.125 as a fraction
= _____

d $\frac{3}{5}$ as a decimal
= _____

e $\frac{7}{20}$ as a decimal
= _____

f $\frac{5}{6}$ as a decimal
= _____

g $\frac{4}{9}$ as a decimal
= _____

h $\frac{3}{11}$ as a decimal
= _____

i Round 5.4271 to 1 decimal place.
_____

j Round 12.3958 to 2 decimal places.
_____

2E Multiplying decimals

MEDIUM · core EXAM Q16a

LEARN IT

1. Ignore the decimal points and multiply the whole numbers.
2. Count the decimal places in the question (add them up).
3. Put that many decimal places in the answer.
4. Estimate to check it makes sense.

WORKED EXAMPLE

0.6 × 0.07: 6 × 7 = 42; 1 + 2 = 3 decimal places → **0.042**
2.3 × 1.4: 23 × 14 = 322; 2 decimal places → **3.22** (estimate 2 × 1.5 = 3 ✓)

YOUR TURN 5

a 0.4 × 0.3
4 × 3 = ____
decimal places = ____
answer = ____

b 0.9 × 0.06
= _____

c 2.5 × 0.8
= _____

d 1.6 × 0.45
= _____

e 3.2 × 0.15
= _____

f 7.5 × 0.12
= _____

2F Dividing by a decimal

MEDIUM · core EXAM Q16b

LEARN IT

Move the decimal point in **both** numbers the same number of places to the right, until you are dividing by a **whole number**. (This multiplies both by 10, 100, … so the answer does not change.)

WORKED EXAMPLE

4.2 ÷ 0.07 → × 100 → 420 ÷ 7 = **60**
2.88 ÷ 0.4 → × 10 → 28.8 ÷ 4 = **7.2** (check: 7.2 × 0.4 = 2.88 ✓)

YOUR TURN 6

a 3.6 ÷ 0.4
= ____ ÷ ____
= _____

b 1.44 ÷ 0.12
= _____

c 8.1 ÷ 0.09
= _____

d 0.56 ÷ 0.08
= _____

e 15 ÷ 0.5
= _____

f 9.45 ÷ 0.3
= _____

2G Decimals in real life

CHALLENGING · core, exam style

YOUR TURN 7

Show your working and answer in a sentence.

a A netball club orders 12 pizzas at $14.90 each and 8 garlic breads at $4.75 each. What is the total cost?

b A 1.5 kg bag of rice costs $4.20. What is the cost per kilogram?

c Ella has $50. She buys 3 drinks at $2.85 each and a sandwich for $8.40. How much change does she get?

2H Converting between fractions, decimals and percentages

EASY · support & core

LEARN IT

Per cent means "out of 100".

- **% → fraction:** write it over 100, then simplify: 35% = $\frac{35}{100}$ = $\frac{7}{20}$
- **% → decimal:** divide by 100: 35% = 0.35 · 4% = 0.04
- **Decimal → %:** multiply by 100: 0.6 = 60%
- **Fraction → %:** multiply by 100%: $\frac{3}{8}$ × 100% = 37.5%

YOUR TURN 8

Complete the table.

Fraction (simplest form)	Decimal	Percentage
a $\frac{1}{4}$		
b	0.6	
c		12%
d $\frac{7}{8}$		
e		2.5%

2I Percentage of a quantity

MEDIUM · core · EXAM Q9a

LEARN IT

"**of**" means ×. Change the percentage to a decimal and multiply: 15% of 80 = 0.15 × 80 = 12.
Without a calculator: 10% → ÷ 10 · 5% → half of 10% · 1% → ÷ 100 · 25% → ÷ 4 · 50% → ÷ 2

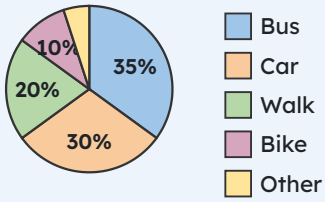
WORKED EXAMPLE

35% of 240: 10% = 24, so 30% = 3 × 24 = 72
5% = half of 24 = 12, so 35% = 72 + 12 = **84**
Calculator: 0.35 × 240 = 84 ✓

YOUR TURN 9

a 10% of 70 = ____ **b** 25% of 36 = ____ **c** 40% of 85 = ____ **d** 12% of $250 = ____ **e** $\frac{3}{8}$ of 64 = ____

160 Year 8 students were asked how they travel to school.



f How many students travel by bus?

____ students

g How many more students walk than ride a bike?

____ more students

2J One quantity as a percentage of another

MEDIUM · core

LEARN IT

Write the amounts as a fraction, **part over whole**, then multiply by 100%. Make sure both amounts are in the **same units** first.

WORKED EXAMPLE

18 out of 24 = $\frac{18}{24}$ × 100% = **75%**
450 m out of 2 km = $\frac{450}{2000}$ × 100% = **22.5%** (2 km = 2000 m)

YOUR TURN 10

a Chloe scored 34 out of 40 in a quiz. What percentage is this?

$\frac{\quad}{\quad}$ × 100% = ____

b A netballer had 30 shots at goal and scored 12. What percentage of her shots went in?

____

c What percentage of 2 hours is 30 minutes? Make the units the same first.

____

2K Percentage increase and decrease

CHALLENGING · core, exam style

LEARN IT

Increase: find the percentage of the amount, then **add** it on.
Decrease (or discount): find the percentage, then **subtract** it.

WORKED EXAMPLE

Increase $80 by 15%: 15% of 80 = 12 → 80 + 12 = **$92**
$120 shoes, 25% off: 25% of 120 = 30 → 120 − 30 = **$90**

YOUR TURN 11

a Increase $60 by 10%.

10% of 60 = ____
60 + ____ = ____

b Decrease 250 kg by 20%.

____ kg

c A $45 T-shirt is on sale at 30% off. What is the sale price?

____

d 30% of the students in Year 8 is 54 students. How many students are in Year 8? Hint: find 1% first.

____ students

I can do this on my own: ☐ fractions ☐ F ↔ D ↔ % ☐ × and ÷ decimals ☐ % of an amount ☐ % increase and decrease

Check your answers in the answer booklet. Diagrams are drawn to scale unless marked "not to scale".

3 Algebraic techniques

NSW syllabus: **MA4-ALG-C-01** generalises number properties to operate with algebraic expressions including expansion and factorisation · **MAO-WM-01** working mathematically

Yearly exam
Section C · Q19

3A The language of algebra

EASY · support & core

LEARN IT

$5x - 3y + 7$

- x and y are **pronumerals**: letters that stand for numbers.
- There are 3 **terms**: $5x$, $-3y$ and 7. The sign in front belongs to the term.
- 5 is the **coefficient** of x (the number in front). The coefficient of y is -3.
- 7 is the **constant term** (a number on its own).

LEARN IT

How we write algebra

$3 \times a = 3a$ · $a \times b = ab$ · $b \times a = ab$ (alphabetical)

$a \times a = a^2$ · $1 \times x = x$ · $a \div 4 = \frac{a}{4}$

Expanded form shows every ×: $5ab = 5 \times a \times b$

YOUR TURN 1

a For $6k - p + 4$, write: (i) the coefficient of k ____ (ii) the coefficient of p ____ (iii) the constant term ____ (iv) the number of terms ____

Write simply (b–f), then **write in expanded form** (g–h).

b $5 \times y =$ ____ **c** $a \times 3 \times b =$ ____ **d** $m \times m =$ ____ **e** $2 \times p \times p =$ ____ **f** $k \div 6 =$ ____

g $4mn =$ ____ **h** $3c^2 =$ ____

3B Turning words into algebra

EASY · core

LEARN IT

+ sum, more than, increased by, plus
− difference, less than, decreased by
× product, times, twice (× 2), double
÷ quotient, divided by, shared, half (÷ 2)

Careful: "3 less than n" is $n - 3$, not $3 - n$.

YOUR TURN 2

Write an expression for each. Use the pronumerals given.

a 7 more than n

b the product of 4 and n

c 3 less than n

d n divided by 5

e twice n, then add 1

f the next whole number after n

g the sum of x and y, all multiplied by 3

3C Substitution

MEDIUM · core EXAM Q19c

LEARN IT

Replace each pronumeral with its number **in brackets**, then use the order of operations:
brackets → powers → × and ÷ → + and −
$3a^2$ means $3 \times a^2$: square first, then multiply. It is **not** $(3a)^2$.

WORKED EXAMPLE

When $a = 4$ and $b = -2$:

$5a - 3 = 5 \times (4) - 3 = 20 - 3 = \mathbf{17}$

$3a^2 + b = 3 \times (4)^2 + (-2) = 3 \times 16 - 2 = 48 - 2 = \mathbf{46}$

$b^2 = (-2)^2 = (-2) \times (-2) = \mathbf{4}$

YOUR TURN 3

Evaluate each expression when $x = 5$, $y = 2$ and $z = -3$.

a $3x + y$
$3 \times (5) + (2)$
$=$ ____ $+$ ____ $=$ ____

b $xy - 4$
$= (5) \times ($____$) - 4$
$=$ ____

c $x^2 + y^2$
$=$ ____

d $2y^2$
$2 \times (2)^2$
$= 2 \times$ ____ $=$ ____

e $3x^2 - 4y$
$=$ ____

f $x + z$
$=$ ____

g z^2
$=$ ____

h $4z - y$
$=$ ____

i $(x + y)^2$
$=$ ____

j An electrician charges $C = 45h + 60$ dollars for a job that takes h hours. Find C when (i) $h = 3$ (ii) $h = 2.5$

3D Collecting like terms

EASY · support & core · EXAM Q19a

LEARN IT

Like terms have exactly the same pronumerals, with the same powers.
Like: $3x$ and $-5x$ · $4ab$ and $2ba$ (ab = ba)
Not like: $2x$ and $2x^2$ · $3x$ and $3y$
To simplify, **add or subtract the coefficients** of like terms. Keep the pronumeral the same. The sign **in front** of a term moves with it.

WORKED EXAMPLE

$7m + 3n - 2m + n$
$= 7m - 2m + 3n + n$ ← group like terms
$= \mathbf{5m + 4n}$ (n means $1n$)
$4x^2 + 3x - x^2 + 5x = \mathbf{3x^2 + 8x}$
$2k - 7 - 5k + 4 = \mathbf{-3k - 3}$

YOUR TURN 4

Simplify by collecting like terms.

a $6a + 2a$
$= (__ + __)a = ____$

b $9k - k$
$= ____$

c $3p + 4 + 2p$
$= ____$

d $5x + 3y + 2x - y$
$= ____$

e $8p - 3q - 5p + 6q$
$= ____$

f $3ab + 2ba - ab$
$= ____$

g Circle the terms that are like $5x$: $2y$ $-x$ x^2 $7x$ 5

h $6y^2 - 2y + y^2 + 5y$
$= ____$

i $2m - 7 - 5m + 4$
$= ____$

j Simplify the perimeter, P, of this rectangle. (not to scale)



$P = ____$

3E Multiplying terms

MEDIUM · core

LEARN IT

Multiply the **numbers**, then the **pronumerals** (in alphabetical order). The same letter twice is squared.
Signs: + × + = + · + × − = − · − × − = +

WORKED EXAMPLE

$3a \times 4b = 3 \times 4 \times a \times b = \mathbf{12ab}$ $2x \times 5x = \mathbf{10x^2}$
$-3m \times 4n = \mathbf{-12mn}$ $-2p \times (-6q) = \mathbf{12pq}$

YOUR TURN 5

Simplify.

a $5 \times 3k$
$= __ \times __ \times k$
$= ____$

b $2p \times 7q$
$= ____$

c $4y \times 3y$
$= ____$

d $-2a \times 6b$
$= ____$

e $3m \times 2n \times 5$
$= ____$

f $-4x \times (-3x)$
$= ____$

3F Dividing terms

MEDIUM · core

LEARN IT

Write the division as a **fraction**. Then cancel factors that are on the top **and** the bottom: numbers first, then pronumerals.

WORKED EXAMPLE

$12xy \div 4x = \frac{12xy}{4x} = \mathbf{3y}$ ($12 \div 4 = 3$ and $x \div x = 1$)
$\frac{20a^2}{5a} = \frac{20 \times a \times a}{5 \times a} = \mathbf{4a}$

YOUR TURN 6

Simplify.

a $18m \div 6$
$= \frac{18m}{6} = ____$

b $15xy \div 5y$
$= ____$

c $\frac{24ab}{8a}$
$= ____$

d $\frac{14k^2}{7k}$
$= ____$

e $\frac{30mn}{-6n}$
$= ____$

3G Reasoning with algebra

CHALLENGING · core, exam style

YOUR TURN 7

To test whether two expressions are equal, substitute the same numbers into both. If the answers differ even once, they are not equal. Explain using words such as *like terms* and *square first*.

a Sam says $3x + 2y = 5xy$. Test this with $x = 2$ and $y = 1$. Then explain Sam's mistake.

b Find the missing term:
$7m - ____ = 3m$

c Find the missing term:
$6ab \div ____ = 2b$

d Simplify the area of a rectangle $5x$ long and $3x$ wide:
Area = $____$

e Is $4a^2$ the same as $(4a)^2$? Test with $a = 3$.

3H Expanding brackets (the distributive law)

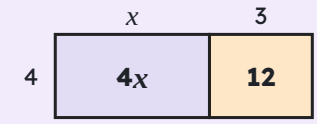
MEDIUM · core · EXAM Q19b

LEARN IT

Multiply the term outside by **every** term inside:

$$a(b + c) = ab + ac$$

A negative outside changes the sign of **every** term inside.



$4(x + 3) = 4x + 12$ (not to scale)

WORKED EXAMPLE

$5(2y - 3) = 5 \times 2y - 5 \times 3 = \mathbf{10y - 15}$

$-3(a + 4) = \mathbf{-3a - 12}$ $\quad x(x + 7) = \mathbf{x^2 + 7x}$

Expand, then simplify: $2(x + 5) + 3x = 2x + 10 + 3x = \mathbf{5x + 10}$

YOUR TURN 8

Expand. Simplify where you can.

a $6(m + 2)$
$= 6 \times m + 6 \times 2$
$=$ _____

b $3(4k - 5)$
$=$ ___ $-$ ___

c $-2(y + 8)$
$=$ _____

d $a(a - 9)$
$=$ _____

e $4(2x + 1) + 3$
$=$ _____

f $5(p + 2) + 2(p - 3)$
$=$ _____

3I Factorising

CHALLENGING · core, exam style

LEARN IT

Factorising is the **reverse** of expanding.

1. Find the **HCF** of all the terms (number and pronumeral).
2. Write the HCF outside the brackets.
3. Divide each term by the HCF. Write what is left inside.
4. Check by expanding.

WORKED EXAMPLE

$6x + 15$: HCF = 3 → $\mathbf{3(2x + 5)}$ check: $3 \times 2x + 3 \times 5$ ✓

$8a - 12 = \mathbf{4(2a - 3)}$ $\quad x^2 + 5x = \mathbf{x(x + 5)}$

$10ab + 15a$: HCF = $5a$ → $\mathbf{5a(2b + 3)}$

YOUR TURN 9

Factorise fully.

a $4x + 8$
HCF = ___
= ___(___ + ___)

b $9m - 6$
$=$ _____

c $12y + 18$
$=$ _____

d $k^2 - 7k$
$=$ _____

e $14pq + 21p$
$=$ _____

MIX Mixed exam check: all three topics · 12 marks · about 12 minutes

CHALLENGING · core, exam style · EXAM SECTION C

YOUR TURN ★

Try these without looking back. Then check, and revise any you got wrong.

1 Simplify $9a + 2b - 4a - 6b =$ _____ 1 mark

2 Expand $4(3m - 2) =$ _____ 1 mark

3 Evaluate $3p^2 - 2q$ when $p = 2$ and $q = -5$. 1 mark

_____

4 Find the hypotenuse of a right-angled triangle with shorter sides 16 cm and 30 cm. 2 marks

_____ cm

5 Calculate $0.7 \times 0.08 =$ _____ 1 mark

6 Calculate $2.4 \div 0.03 =$ _____ 1 mark

7 A sector graph shows that 45% of 360 students play sport. How many students is that? 1 mark

_____ students

8 Evaluate $2\frac{1}{2} + \frac{3}{4}$ 1 mark

_____

9 Is a triangle with sides 7 cm, 9 cm and 12 cm right-angled? Show working. 2 marks

10 Increase \$150 by 8%. 1 mark

_____

I can do this on my own: ☐ substitute ☐ like terms ☐ × and ÷ terms ☐ expand ☐ factorise